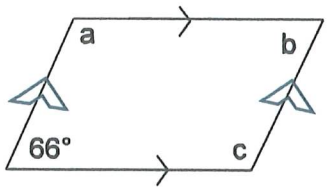
Calculate the following angles and or lengths, giving reasons:

1.

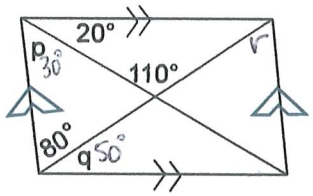


a = $180° - 66° = 114°$ (co-int ∠'s, // lines)

b = $66°$ (opp ∠'s parm)

c = $114°$ (opp ∠'s parm)

2.

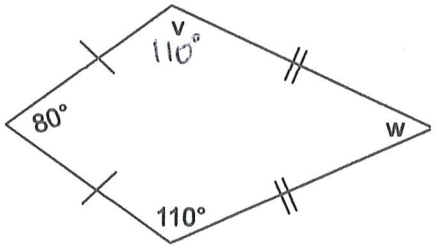


p = $110° - 80° = 30°$ (ext ∠ △)

q = $180° - (30° + 20° + 80°) = 50°$ (co-int ∠'s // lines)

r = $80°$ (alt ∠'s, // lines)

3.

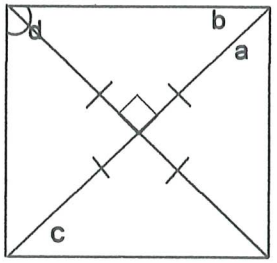


v = $110°$ (opp ∠'s kite)

w = $360° - (110° + 110° + 80°)$ (∠ sum quad)
$= 60°$

4. Why is the shape below a square?

equal diagonals bisect at 90°



a = $45°$ (diag sq bisects ∠)

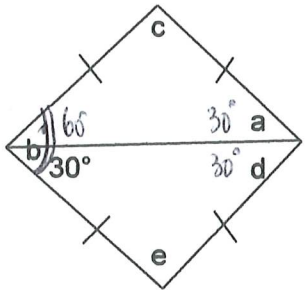
b = $45°$ (")

c = $45°$ "

d = $90°$ (∠'s of a square)

5. **What shape is the shape below and why?**

Rhombus - 4 equal sides, no 90°



a = 30° (alt ∠'s, sides of rhombus //)

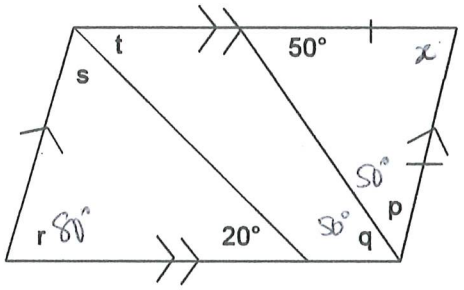
b = 60° (30° = 30° ∠'s opp = sides; 30° + 30° = 60°)

c = 120° (∠ sum △)

d = 30° (∠'s opp = sides)

e = 120° (∠ sum △)

6.



p = 50° (∠'s opp = sides)

q = 50° (alt ∠'s, // sides)

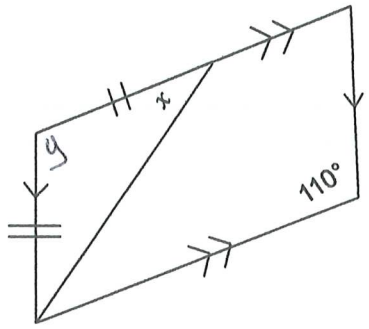
x = 80° (∠ sum △)

r = 80° (opp ∠'s parm)

s = 80° (∠ sum △)

t = 20° (alt ∠'s // lines)

7.

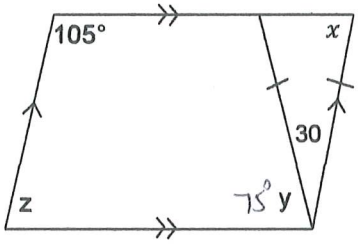


$y = 110°$ (opp L's parm =)

$x = \frac{180° - 110°}{2}$ (L sum isos Δ)

$x =$ 35°

8.



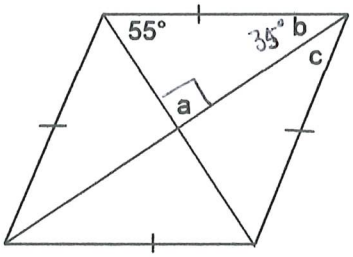
(L sum isos Δ) OR..)..

$x =$ 75° (co-int L's, // lines)

$y =$ 105° − 30° = 75° (opp L's parm =) OR alt L's // lines

$z =$ x = 75° (opp L's parm =)

9.

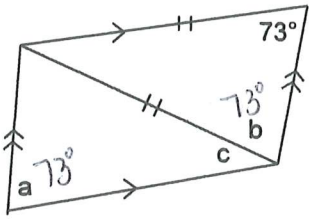


a = 90° (diago rhombus)

b = 90° − 55° = 35° (L sum Δ)

c = 35° (diago rhombus, bisect L's)

10.



a = 73° (opp L's parm)

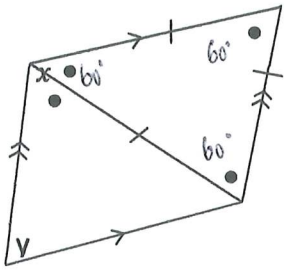
b = 73° (L's opp = sides Δ)

c = 180° − (73° + 73°) (co-int L's, // lines)

= 180° − 146°

= 34°

11.

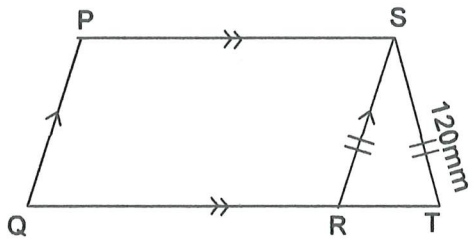


• = 60° (∠ sum △)

x = 2• = 120°

y = 60° (opp ∠'s parm =)

12. Calculate the following lengths, giving reasons



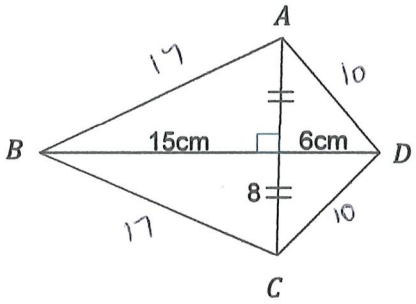
PQ ?

ST = SR = 120mm (= sides, given)

SR = PQ = 120mm

(opp sides parm =)

13.



Perimeter $ABCD$?

CD = $\sqrt{8^2+6^2}$ = 10cm (Pythag)

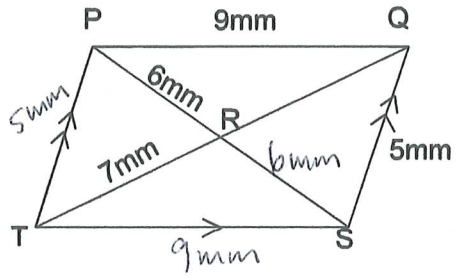
AD = CD = 10cm (short sides of kite =)

BC = $\sqrt{15^2+8^2}$ = 17cm (Pythag)

AB = BC = 17cm (long sides kite =)

Perimeter = 54 cm

14.



TS = 9mm (opp sides parm =)

PT = 5mm (")

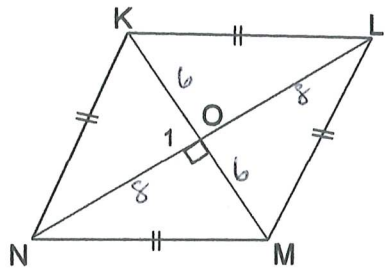
RS = 6mm (diags parm bisect)

PS = 12mm

15. What shape is KLMN? Rhombus

Why? 4 equal sides

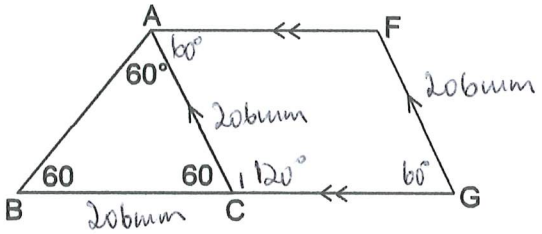
What size is $\hat{O}_1$? 90° (adj L's str line) OR (diags rhombus bisect at 90°)



KM = 12 units, NL = 16 units.
Find the perimeter of KLMN.

Each diagonal is bisected in a rhombus.
∴ KO = OM = 6 and NO = OL = 8
NM = 10 (Pythag)
∴ Perimeter KLMN = 40 units

16.



a) FG = 206mm, what is the length of BC? 206mm opp sides parm and = sides equilateral △.

b) Give the sizes of the missing angles.

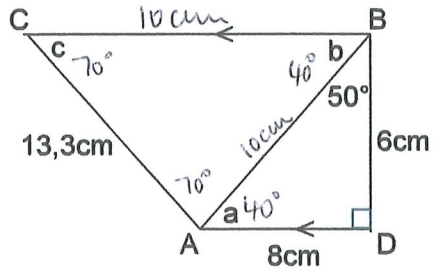
$\hat{FAC} = 60°$ (alt L's AF // CG)

$\hat{A} = 60°$ (opp L's parm)

$\hat{C}_1 = 120°$ (adj L's str line)

$\hat{F} = 120°$ (opp L's parm)

17. Find the following giving reasons:



AB in BC = 10cm.

AB = 10cm (Pythag)

∴ △ ABC is isos, BC = AB.

a) The size of angles a, b and c

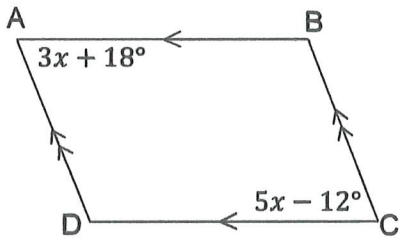
$a = 40°$ (L sum △)

$b = 40°$ (alt L's, CB // AD)

$C = \frac{180° - 40°}{2} = 70°$ (L sum isos △)

b) The perimeter of ADBC if BC = 10cm

18. Find the sizes of the angles of ABCD, giving reasons:



$3x + 18° = 5x - 12°$ (opp L's parm =)

$30° = 2x$

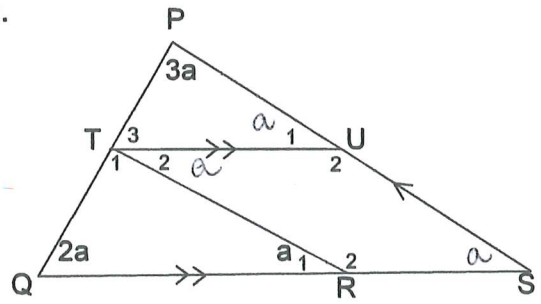
$15° = x$

$\hat{A} = 3(15°) + 18° = 63°$

$\hat{C} = 63°$

$\hat{D} = \hat{B} = 180° - 63° = 117°$ (co-int L's // lines)

19. Find the value of a, giving reasons.



$\hat{T}_2 = \hat{R}_1 = a$ (alt L's, TU // QS)

$\hat{U}_1 = a$ (alt L's, PS // TR)

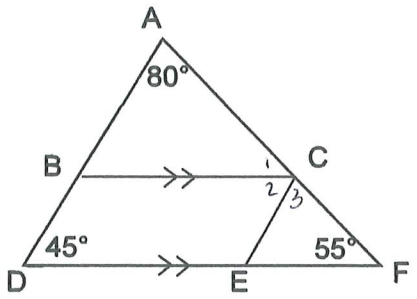
$\hat{S} = a$ (corresp L's, TU // QS OR opp L's parm)

$3a + 2a + a = 180°$ (L sum △PQS)

$6a = 180°$

$a = 30°$.

20.

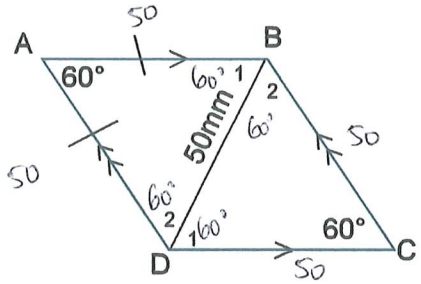


Find the angles of parm BCED.

$\hat{C}_2 = 45°$ (opp ∠'s parm)

$\hat{B} = \hat{E} = 135°$ (co-int ∠'s // lines)

21.



a) Why is ABCD a rhombus?

$\hat{B}_1 = \hat{D}_2 = 60°$ (∠ sum isos Δ ABD)

$\therefore \hat{D}_1 = \hat{B}_2 = 60°$ (alt ∠'s // lines)

∴ Both Δ's are equilateral, all sides = 50mm

∴ ABCD rhombus (both prs opp sides =)

b) Find the size of $\angle B_1$

found above

c) Calculate the perimeter of ABCD.

$P = 4(50) = 200$ mm.